

Lesson 2.3: Group Isomorphism

Learning Outcomes: At the end of the lesson, the students are able to:

- test if a given map is a group isomorphism or not;
- construct isomorphisms in order to show that two groups are isomorphic.
- derive properties of group isomorphisms.

Isomorphic Groups

Let $G = \{I, R, A, L\}$, where the elements are military training commands given as follows:

- I : “Attention.”
- R : “Right face.”
- A : “About face.”
- L : “Left face”

Let $\circ$ denote composition of commands, that is, if $X, Y \in G$, then $X \circ Y$ means the result when we perform command Y , followed by the command X . For example,

$$R \circ A = L$$

since an “about face” command, followed by a “right face” simplifies into a “left face” command. See Figure 7.

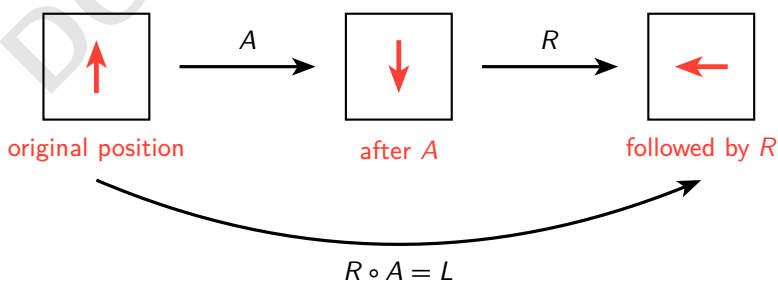


Figure 7: Composition of Commands



By computing all compositions in a similar way, we come up with the Cayley table for G . See Table 10.

$\circ$	I	R	A	L
I	I	R	A	L
R	R	A	L	I
A	A	L	I	R
L	L	I	R	A

Table 10: Cayley Table for G

One checks that $G = \{I, R, A, L\}$ is a group under composition, with identity I . Observe that this Cayley table resembles the Cayley table for $\mathbb{Z}_4$ (Table 6), in a way that if we color corresponding elements, we see the same *color pattern*.

$\circ$	I	R	A	L
I	I	R	A	L
R	R	A	L	I
A	A	L	I	R
L	L	I	R	A

$+$	0	1	2	3
0	0	1	2	3
1	1	2	3	0
2	2	3	0	1
3	3	0	1	2

Table 11: Color Pattern

The same color pattern suggests that, although the groups $(G, \circ)$ and $(\mathbb{Z}_4, +)$ are not the same because their elements are not the same, they are not completely different. Notice that from the group $(G, \circ)$, we may recover the group $(\mathbb{Z}_4, +)$ by replacing the operation $\circ$ by $+$ and do the following *replacements*: $I \mapsto 0$, $R \mapsto 1$, $A \mapsto 2$, $L \mapsto 3$. This idea may be formalized in the following definition.

Definition 2.3.1 (Group Isomorphism)

Let $(G, *)$ and $(H, \blacksquare)$ be groups. A function $\varphi : G \rightarrow H$ is called a **group isomorphism** if the following conditions hold:

1. $\varphi : G \rightarrow H$ is a bijection.
2. $\varphi : G \rightarrow H$ is *operation-preserving* in the sense that

$$\varphi(a * b) = \varphi(a) \blacksquare \varphi(b), \text{ for all } a, b \in G.$$



Note that property (1) in **Definition 2.3.1** requires that G and H must have the same cardinality. Furthermore, functions between groups satisfying property (2) are called *group homomorphisms*, which will be studied in a future lesson. Thus, a bijective group homomorphism is a group isomorphism.

Definition 2.3.2 (Isomorphic Groups)

We say that the group $(G, *)$ is **isomorphic** to the group $(H, \cdot)$, expressed as $(G, *) \cong (H, \cdot)$, if there exists a group isomorphism $\varphi : G \rightarrow H$.

Example 2.3.3 The group $G = \{I, R, A, L\}$ is isomorphic to $\mathbb{Z}_4 = \{0, 1, 2, 3\}$. A group isomorphism necessary to guarantee this is the function $\varphi : G \rightarrow \mathbb{Z}_4$ given by

$$\begin{aligned}\varphi \\ I &\mapsto 0 \\ R &\mapsto 1 \\ A &\mapsto 2 \\ L &\mapsto 3\end{aligned}$$

For instance, $\varphi(R \circ A) = \varphi(L) = 3$ and $\varphi(R) + \varphi(A) = 1 + 2 = 3$, so that

$$\varphi(R \circ A) = \varphi(R) + \varphi(A).$$

The student is asked to check the correctness of other instances.

Example 2.3.4 Let $\mathbb{R}^+$ denote the set of all positive real numbers. As one may check, $\mathbb{R}^+$ forms a group under real number multiplication. We prove that $(\mathbb{R}^+, \cdot) \cong (\mathbb{R}, +)$ by exhibiting a group isomorphism between these groups.

Define $f : \mathbb{R}^+ \rightarrow \mathbb{R}$ as $f(x) = \ln x$, for all $x \in \mathbb{R}^+$. This mapping is well-defined because the natural domain of the natural logarithm is $\mathbb{R}^+$.

From calculus, we know that $f(x) = \ln x$ is a strictly increasing continuous function on the interval $(0, \infty)$, whose range is $(-\infty, \infty)$. Therefore, $f : \mathbb{R}^+ \rightarrow \mathbb{R}$ is a bijection.

We now show that $f : \mathbb{R}^+ \rightarrow \mathbb{R}$ is operation-preserving. Let $a, b \in \mathbb{R}^+$. Then from a well-known property of logarithms, we see that

$$f(a \cdot b) = \ln(ab) = \ln a + \ln b = f(a) + f(b).$$

Therefore, $f : \mathbb{R}^+ \rightarrow \mathbb{R}$ is a group isomorphism and $(\mathbb{R}^+, \cdot) \cong (\mathbb{R}, +)$.



The student is asked to prove the statements in the following remark.

Remark 2.3.5 Let G, H , and K be groups.

1. The identity map $\text{id}_G : G \rightarrow G$ is a group isomorphism.
2. If the function $f : G \rightarrow H$ is a group isomorphism, then $f^{-1} : H \rightarrow G$ is also a group isomorphism.
3. If the functions $f : G \rightarrow H$ and $g : H \rightarrow K$ are group isomorphisms, then $g \circ f : G \rightarrow K$ is also a group isomorphism.

Remark 2.3.5 allows us to see that the relation $\cong$ among the set of all groups is an *equivalence relation*. Meaning, if G, H , and K are groups, then

1. $G \cong G$;
2. $G \cong H$ implies $H \cong G$; and
3. $G \cong H$ and $H \cong K$, implies $G \cong K$.

The equivalence classes defined by this equivalence relation are called the *group isomorphism classes*. In group theory, one of the main goals is to identify all the different group isomorphism classes.

We now formulate necessary conditions for a function to be a group isomorphism. These conditions will later allow us to tell when two groups are not isomorphic.

Theorem 2.3.6 Let $\varphi : G \rightarrow H$ be a group isomorphism.

1. If e_G and e_H respectively denote the identity elements of G and H , then $\varphi(e_G) = e_H$.
2. For all $a \in G$, $\varphi(a^{-1}) = \varphi(a)^{-1}$.
3. For all $n \in \mathbb{Z}$, $\varphi(a^n) = \varphi(a)^n$.
4. For all $a \in G$, $|a|_G = |\varphi(a)|_H$.
5. For all $a, b \in G$, the elements a and b commute in G if and only if $\varphi(a)$ and $\varphi(b)$ commute in H .
6. The group G is abelian if and only if the group H is abelian.



Proof.

1. Note that

$$f(e_G)f(e_G) = f(e_G e_G) = f(e_G) = f(e_G) e_H.$$

By the left cancellation law in H , it follows that $f(e_G) = e_H$.

2. Let $a \in G$. Then

$$f(a)f(a^{-1}) = f(aa^{-1}) = f(e_G) = e_H.$$

In a similar way, $f(a^{-1})f(a) = e_H$. Therefore, $f(a)^{-1} = f(a^{-1})$.

3. Let $a \in G$ and let $n \in \mathbb{Z}$. If $n = 0$, $f(a^n) = f(a)^n$ trivially holds by statement (1). If $n = 1$, then $f(a^n) = f(a) = f(a)^1 = f(a)^n$. Suppose that $f(a^k) = f(a)^k$ holds for some positive integer k . Then

$$f(a^{k+1}) = f(a^k a) = f(a^k)f(a) = f(a)^k f(a) = f(a)^{k+1}.$$

By the principle of mathematical induction, it follows that $f(a^n) = f(a)^n$ holds for all integers $n \geq 0$.

Let n be a negative integer, say $n = -m$ for some positive integer m . By statement (2) and the positive integer case, it follows that

$$f(a^n) = f(a^{-m}) = f((a^m)^{-1}) = f(a^m)^{-1} = f(a)^{-m} = f(a)^n.$$

Therefore, $f(a^n) = f(a)^n$ holds for all $n \in \mathbb{Z}$.

4. This follows directly from statement (3). The student is asked to write the details.
5. Let $a, b \in G$. Then

$$\begin{aligned} ab = ba &\iff ab(ba)^{-1} = e_G \\ &\iff aba^{-1}b^{-1} = e_G \\ &\stackrel{(1)}{\iff} f(aba^{-1}b^{-1}) = e_H \\ &\stackrel{(2)}{\iff} f(a)f(b)f(a)^{-1}f(b)^{-1} = e_H \\ &\iff f(a)f(b) = f(b)f(a). \end{aligned}$$



6. This follows directly from statement (5). $\square$

As a consequence of **Theorem 2.3.6**, the following remarks may now be stated as facts.

Remark 2.3.7 If any of the statements below is true, then G is **not** isomorphic to H .

1. G and H have different cardinalities.
2. G has exactly m_1 elements of order k and H has exactly m_2 elements of order k , but $m_1 \neq m_2$.
3. G is abelian, but H is non-abelian.

Example 2.3.8 The following pairs of groups are **not** isomorphic.

1. $(\mathbb{Z}, +)$ and $(\mathbb{R}, +)$

This is because $\mathbb{R}$ and $\mathbb{Z}$ have different cardinalities as infinite sets. It is a fact from **Math 105** that there does not exist a surjective function $f : \mathbb{Z} \rightarrow \mathbb{R}$.

2. $(\mathbb{Z}_4, +)$ and $(V_4, \circ)$

The group $\mathbb{Z}_4$ has only one element of order 2, while V_4 has exactly three elements of order 2 (as the student will check)

3. $(\mathbb{Z}_6, +)$ and $(S_3, \circ)$

The group $(\mathbb{Z}_6, +)$ is abelian, while $(S_3, \circ)$ is non-abelian.

Example 2.3.8 shows that for a given positive integer n , there may exist two groups of order n which are not necessarily isomorphic. In finite group theory, one of our primary goal is to classify groups of a given order n , *up to isomorphism*. This means that we would like to find out how many group isomorphism classes are there for a given order. As we shall shortly find out, this is not a trivial task.

For instance, $\mathbb{Z}_4$ and V_4 are non-isomorphic groups of order 4. Therefore, there are at least two groups of order 4, up to isomorphism. The question remains whether there are any others.

